UNIFICATION AND RESOLUTION

Aim

To implement and demonstrate **Unification and Resolution in Prolog** using sentence matching, compound term unification, and conflict resolution.

Exercise 1: Sentence Matcher

% Facts

sentence(the_cat_sits).

sentence(the_dog_runs).

% Query

% ?- sentence(X).

Output:

X = the_cat_sits ; X = the_dog_runs ; false.

Exercise 2: Compound Term Unification

% Facts

likes(john, food(pizza, spicy)).

likes(mary, food(pizza, sweet)).

% Query

% ?- likes(john, food(pizza, T)).

Output:

T = spicy.

Exercise 3: Conflict Resolution

```
% Facts

rule1(climate_change).

rule2(climate_change).

% Rule

conflict(X) :- rule1(X), rule2(X).

% Query

% ?- conflict(climate_change).
```

Output:

```
true.
```

Result

Thus, the **Unification and Resolution** concepts were successfully implemented in Prolog. The queries were executed successfully and the expected results were obtained.